



英検2級 大問2B 長文空所補充 完全演習

全30セット / 問題編

THINKING 大問別問題集シリーズ | 本番そっくりの形式で30題を演習。

- ・各長文に空所が3つ。文意に沿って **1, 2, 3, 4** から最適なものを選ぶ。
 - ・**コツ**：空所は**選択肢を見る前に**「どんな内容（+/-）が入るか」を前後の論理・接続語から予測 → 選択肢と照合。
 - ・解いたら巻末の解答で答え合わせ → **解説編**で「なぜ正解／なぜ誤りか」と本質的な読み方を確認。
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B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.1 / あくびが移る理由

Why Yawning Spreads

Everyone yawns, but scientists are still not completely sure why. A yawn happens when a person opens the mouth wide and takes a deep breath. One idea is that yawning (1). When we are tired or bored, the brain may get a little too warm, and a deep breath of cool air helps bring its temperature back down. This may be one reason why we yawn more often late at night or in a stuffy, crowded room.

What is even more interesting is that yawning is "contagious." When people see or hear someone else yawn—or even read about yawning—they often feel the need to yawn too. Scientists believe this happens because (2). Humans are social animals, and copying others' actions without thinking may help groups feel connected and stay alert together.

Interestingly, contagious yawning is stronger between people who are close, such as family members and good friends. (3), the more we care about someone, the more likely we are to "catch" their yawn. Studying this simple action may help scientists understand how humans connect with one another.

- (1)
- 1 helps cool the brain
 - 2 makes people fall asleep
 - 3 cleans the teeth
 - 4 stops people from breathing

- (2)
- 1 yawning is bad for health
 - 2 reading makes people tired
 - 3 our brains naturally copy what others do
 - 4 people are simply being polite

- (3)
- | | |
|---------------|------------------|
| 1 By contrast | 2 In other words |
| 3 Until then | 4 However |

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No.2 / ミツバチの会話

How Bees Talk

When a honeybee finds a good source of food, it needs to tell the other bees in the hive where it is. Bees cannot speak, so they share this information in a surprising way: (1). By moving its body in a special pattern, often called a "waggle dance," a single bee can describe the exact direction and distance of the flowers it has found.

The dance carries a lot of information. The direction in which the bee moves shows the direction of the food compared to the sun. The length of the dance shows (2). A longer dance means the food is farther away, while a shorter one means it is close to the hive. The bee may also repeat the dance many times so that the message is clear to everyone nearby.

Other bees watch the dance closely and then fly straight to the food. This clever system allows the whole hive to gather food quickly and efficiently. (3), the waggle dance is one of the most amazing examples of communication in the animal world.

- | | | | | |
|-----|---|-----------------------|---|-------------------|
| (1) | 1 | by making loud sounds | 2 | by changing color |
| | 3 | by drawing maps | 4 | through dance |

- (2)
- 1 how dangerous the trip is
 - 2 how far the food is
 - 3 how old the bee is
 - 4 how cold the weather is

- | | | | | |
|-----|---|-------------|---|------------|
| (3) | 1 | In fact | 2 | However |
| | 3 | By contrast | 4 | Until then |

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No.3 / コウモリの反響定位

How Bats See with Sound

Bats are active at night, when there is little or no light. To find their way and catch insects in the dark, they do not rely on their eyes. Instead, (1). A bat sends out high sounds that humans cannot hear, and these sounds travel through the air until they hit an object.

When the sound hits an insect or a wall, it bounces back to the bat as an echo. By listening to these echoes, (2). The bat can tell how far away an object is, how big it is, and even which way it is moving—all in complete darkness.

This skill, called echolocation, is so accurate that bats can catch tiny insects while flying at high speed. Scientists have studied it to improve technology such as systems that help ships and submarines find objects underwater. (3), nature has taught humans how to find things using sound.

- | | | | | |
|-----|---|----------------------|---|------------------|
| (1) | 1 | they sleep all night | 2 | they use sound |
| | 3 | they make light | 4 | they stop flying |

- | | | |
|-----|---|--|
| (2) | 1 | the insect escapes easily |
| | 2 | the bat becomes confused |
| | 3 | the sound disappears |
| | 4 | the bat builds a picture of its surroundings |

- | | | | | |
|-----|---|-------------|---|-------------|
| (3) | 1 | However | 2 | By contrast |
| | 3 | In this way | 4 | Until then |

2

B

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No.4 / ヤモリが壁を登る仕組み

How Geckos Climb Walls

Geckos are small lizards that can run up smooth walls and even walk across ceilings without falling. For a long time, people wondered how they could do this. Some thought the geckos used a kind of glue or tiny hooks, but (1). The real secret lies in the special structure of their feet.

The bottom of a gecko's foot is covered with millions of tiny hairs, far too small to see. These hairs are so close to the surface of the wall that (2). This weak force, multiplied across millions of hairs, is strong enough to hold the whole gecko in place.

Scientists have copied this idea to design new kinds of sticky tape and gloves that can grip smooth surfaces without leaving any mark. (3), studying a small lizard's feet has led to useful inventions. Learning from nature in this way is becoming more and more common in science.

- (1)
- 1 both ideas were true
 - 2 the geckos used glue
 - 3 neither idea was correct
 - 4 the walls were sticky

- (2)
- 1 they create a tiny pulling force
 - 2 they push the gecko away
 - 3 they make the wall dirty
 - 4 they produce a loud sound

- (3)
- | | |
|--------------|---------------|
| 1 However | 2 By contrast |
| 3 Until then | 4 In this way |

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No.5 / 木が栄養を分け合う

How Trees Share Food

A forest may look like a group of trees standing alone, but under the ground, something surprising is happening. The roots of many trees are connected by thin threads of fungus. Through this hidden network, (1). Trees can actually send food, water, and even messages to one another.

Scientists have found that older, taller trees often share their extra sugar with younger trees that do not get enough sunlight. (2), if one tree is attacked by insects, it can send a warning through the network so that nearby trees can prepare to defend themselves.

Because of this underground system, a forest acts almost like a single living thing rather than a group of separate trees. Some scientists even call it the "wood wide web." Studying it has changed (3). People now understand that trees are far more connected and cooperative than anyone once believed.

- (1)
- 1 the trees are linked together
 - 2 the trees grow apart
 - 3 the soil becomes dry
 - 4 the fungus dies quickly

- (2)
- | | |
|---------------|------------|
| 1 By contrast | 2 However |
| 3 Until then | 4 Moreover |

- (3)
- 1 the color of the leaves
 - 2 how we think about forests
 - 3 the price of wood
 - 4 the speed of the wind

2

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No.6 / 渡り鳥のナビ

How Birds Find Their Way

Every year, many birds fly thousands of kilometers between their summer and winter homes. They can find the right path even over wide oceans, where there are no roads, signs, or other landmarks to guide them. The journey may last for many days and nights, and the weather is not always clear. (1), they do not get lost. Scientists have discovered that birds use several clever methods to find their way.

One method uses the sun and stars. During the day, birds use the position of the sun in the sky, and at night, they read the patterns of the stars to keep moving in the correct direction. Another method is even more surprising: many birds (2). They can sense the Earth's magnetic field and use it like a built-in compass to know which direction is north, even when the sky is completely dark.

By combining these methods, birds can travel huge distances with great accuracy and arrive at almost the same place year after year. Even today, scientists (3). They are still trying to understand exactly how a bird can sense the Earth's magnetic field inside its own body.

- | | | | | |
|-----|---|-------------|---|-------------|
| (1) | 1 | By contrast | 2 | Even so |
| | 3 | Until then | 4 | For example |

- | | | |
|-----|---|--------------------------------|
| (2) | 1 | can hear other birds far away |
| | 2 | can smell the ocean below |
| | 3 | can feel the Earth's magnetism |
| | 4 | can count the days |

- | | | |
|-----|---|--------------------------------------|
| (3) | 1 | do not fully understand this ability |
| | 2 | think birds cannot navigate |
| | 3 | have stopped their research |
| | 4 | believe birds use paper maps |

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No.7 / ペンギンの防寒

How Penguins Stay Warm

Emperor penguins live in Antarctica, one of the coldest places on Earth, where temperatures can drop far below freezing. To survive in such conditions, penguins have several ways of keeping warm. One of the most important is (1). Their bodies are covered with thick layers of feathers and fat that hold in heat and keep out the cold wind.

Penguins also help each other stay warm. On the coldest days, hundreds of penguins stand close together in a large group called a huddle. By doing this, (2). The penguins on the outside block the wind, while those on the inside share their body heat and stay warm.

To be fair, the penguins on the outside of the huddle feel the cold the most. (3), the birds slowly take turns: those on the outside move to the inside, and those who have been warm move to the edge. This teamwork helps the whole group survive the long, freezing winter.

- (1)

1

their loud voices

2

their long legs

3

their sharp eyes

4

their special bodies
- (2)

1

they keep the whole group warmer

2

they make each other colder

3

they push each other away

4

they fall asleep at once
- (3)

1

However

2

By contrast

3

For this reason

4

Until then

2

B

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No.8 / クモの糸の強さ

How Spider Silk Is So Strong

Spider silk is one of the most amazing materials in nature. Although it is extremely thin and light, it is incredibly strong. In fact, a thread of spider silk (1). For its weight, it is stronger than steel, and it can also stretch without breaking. These rare qualities have puzzled scientists for many years.

Spiders make this silk inside their bodies from a special liquid. As the liquid is pulled out through tiny openings, (2). The way the tiny parts of the silk line up gives it both strength and flexibility, allowing a spider to catch insects much larger than itself without tearing its web apart.

Because spider silk is so useful, scientists are trying to make a similar material in laboratories. So far, however, no one has been able to copy it perfectly. If they succeed, it could be used to make stronger ropes, safer car parts, and even medical thread for repairing the body. (3), copying spider silk could lead to many helpful new products.

- (1)
- 1 breaks very easily
 - 2 is heavier than metal
 - 3 is tougher than it looks
 - 4 melts in the sun

- (2)
- 1 it turns into water
 - 2 it hardens into a solid thread
 - 3 it disappears into the air
 - 4 it loses all its strength

- (3)
- | | |
|--------------|---------------|
| 1 However | 2 By contrast |
| 3 Until then | 4 In this way |

2

B

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No.9 / 空が青い理由

Why the Sky Is Blue

Have you ever wondered why the sky looks blue during the day? The answer has to do with sunlight and the air around us. Although sunlight looks white, (1). It is actually made up of many colors mixed together, including red, yellow, green, and blue.

When sunlight enters the air, it hits tiny gas particles and scatters in all directions. Blue light scatters much more than the other colors because it travels in shorter waves. (2), the blue light spreads across the whole sky, which is why we see blue above us during the day.

At sunset, the sunlight has to travel through much more air to reach our eyes. By the time it arrives, most of the blue light has scattered away, and mainly red and orange light is left. This is (3). The same scattering of light that makes the sky blue also paints the beautiful colors of a sunset.

- (1)
- 1 it is not really a single color
 - 2 it has no color at all
 - 3 it is always blue
 - 4 it cannot be seen

- (2)
- | | |
|---------------|---------------|
| 1 However | 2 By contrast |
| 3 As a result | 4 Until then |

- (3)
- 1 why the moon is bright
 - 2 why clouds are white
 - 3 why stars appear at night
 - 4 why sunsets look red and orange

2

B

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No.10 / 氷が浮く理由

Why Ice Floats

For almost every material in the world, the solid form is heavier than the liquid form and sinks in its own liquid. Water, however, is unusual and behaves in the opposite way. When water freezes into ice, (1). This is why ice cubes float in a glass and why ice forms on top of a pond rather than at the bottom.

The reason for this lies in the special way that water particles join together when they freeze. As water turns to ice, the particles (2). Instead of packing tightly, they form an open structure with more space between them, so the same amount of water takes up more room as ice and weighs less for its size.

This strange property is very important for life. In winter, a layer of ice on the surface of a lake acts like a blanket, keeping the water below from freezing solid. (3), fish and other living things can survive the cold winter underneath. A simple fact about water helps protect life in nature.

- (1)
- 1 it becomes much heavier
 - 2 it becomes lighter than the water around it
 - 3 it turns into gas
 - 4 it sinks to the bottom

- (2)
- 1 come much closer together
 - 2 break into pieces
 - 3 disappear completely
 - 4 spread farther apart

- (3)
- | | |
|---------------|--------------|
| 1 As a result | 2 However |
| 3 By contrast | 4 Until then |

2

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No.11 / 体温調整

How the Body Stays at the Right Temperature

The human body works best at a temperature of about 37 degrees Celsius, and it must stay close to this even when the weather is very hot or very cold. (1), the body has clever ways of controlling its own temperature. A part of the brain acts like a thermostat, always checking whether the body is too warm or too cool.

When the body becomes too hot, the brain tells it to sweat. As the sweat dries on the skin, (2). This carries heat away from the body and helps cool it down. At the same time, blood moves closer to the skin so that more heat can escape into the air.

When the body becomes too cold, it does the opposite. The muscles shake quickly in a movement called shivering, which produces heat, and blood moves away from the skin to keep the inside of the body warm. All of these actions (3). They happen automatically, keeping the body's temperature just right even without us noticing.

- | | | | | |
|-----|---|-----------------|---|------------|
| (1) | 1 | By contrast | 2 | However |
| | 3 | For this reason | 4 | Until then |

- | | | |
|-----|---|-----------------------------|
| (2) | 1 | it takes heat from the skin |
| | 2 | it makes the skin hotter |
| | 3 | it stops the blood |
| | 4 | it has no effect |

- | | | |
|-----|---|-------------------------------|
| (3) | 1 | make the body sick |
| | 2 | help control body temperature |
| | 3 | waste the body's energy |
| | 4 | stop the heart |

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No.12 / プラセボ効果

The Placebo Effect

Sometimes, a patient is given a "medicine" that contains no real drug at all—perhaps just a sugar pill. Surprisingly, some of these patients still feel better afterward. This strange result is known as (1). It shows that the mind can have a powerful effect on how the body feels.

Scientists believe this happens because of what the patient expects. If a person strongly believes that a treatment will work, (2). The brain may release natural chemicals that reduce pain or improve mood, even though the pill itself did nothing.

The placebo effect does not mean the pain was not real. Instead, it shows how closely the mind and body are connected. (3), doctors and scientists must be careful when testing new medicines, comparing them with sugar pills to see whether a drug truly works or whether patients just expected it to.

- | | |
|--|--|
| <p>(1)</p> <p>1 a serious illness</p> <p>3 a waste of time</p> | <p>2 a side effect</p> <p>4 the placebo effect</p> |
|--|--|

- (2)
- 1 the medicine becomes stronger
- 2 the body may actually respond
- 3 the patient gets much worse
- 4 nothing ever changes

- | | |
|--|--|
| <p>(3)</p> <p>1 However</p> <p>3 For this reason</p> | <p>2 By contrast</p> <p>4 Until then</p> |
|--|--|

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No.13 / スポットライト効果

The Spotlight Effect

Have you ever worn something you thought looked strange and felt sure that everyone was staring at you? Most people have had this feeling, but the truth is that (1). Others usually notice us far less than we imagine. Psychologists call this mistake the "spotlight effect."

The reason for this is simple. Each of us pays a lot of attention to ourselves, so we assume that other people are doing the same. (2), everyone else is mostly focused on their own lives and worries. They simply do not have the time or interest to notice every small thing about us.

Understanding the spotlight effect can be very freeing. Once people realize that others are not judging them so closely, (3). They often feel more relaxed, worry less about small mistakes, and find it easier to try new things in front of others.

- | | | | | |
|-----|---|-----------------|---|----------------------|
| (1) | 1 | we are wrong | 2 | everyone is watching |
| | 3 | we look strange | 4 | no one is kind |

- | | | | | |
|-----|---|-------------|---|------------|
| (2) | 1 | For example | 2 | Therefore |
| | 3 | Until then | 4 | In reality |

- | | | |
|-----|---|---------------------------------|
| (3) | 1 | they stop caring about anything |
| | 2 | they feel more afraid |
| | 3 | they become less nervous |
| | 4 | they avoid all people |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.14 / 音楽が心を動かす理由

Why Music Moves Us

Music has a powerful effect on how we feel. A fast, bright song can make us want to dance, while a slow, quiet one can make us feel calm or even sad. But why does (1)? Scientists have been studying how the brain reacts to music to find the answer.

When we listen to music we enjoy, the brain releases a chemical that gives us a feeling of pleasure—the same chemical released when we eat delicious food or receive good news. (2), music can also bring back strong memories. A song from the past may suddenly remind us of a person or a place, along with the feelings we had at the time.

Because music affects both our emotions and our memories, it is used in many ways in daily life. Shops play it to make customers feel relaxed, and films use it to make scenes more exciting or moving. Music even (3). Doctors sometimes use it to help patients feel calmer and recover more comfortably.

- (1)
- 1 music cost so much
 - 2 music change our emotions
 - 3 music sound the same
 - 4 music stop the brain

- (2)
- | | |
|---------------|--------------|
| 1 In addition | 2 However |
| 3 By contrast | 4 Until then |

- (3)
- 1 makes people forget everything
 - 2 has no real use
 - 3 harms our health
 - 4 plays a role in medicine

2

B

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No.15 / 人が忘れる理由

Why We Forget

People often worry when they forget things, but forgetting is a normal and even useful part of how the brain works. Every day, we take in a huge amount of information. (1), the brain cannot and should not keep all of it. Forgetting unimportant details helps make room for the things that really matter.

Memories are not stored like files in a computer. Instead, each time we remember something, the brain rebuilds it. If a memory is not used for a long time, (2). The connections in the brain that hold it grow weaker, and the memory slowly fades until it becomes hard to recall.

This is why repeating and using information helps us remember it. Students who review their notes regularly keep the connections in the brain strong, while those who never review them (3). The lesson is simple: to remember something well, we should use it often. Forgetting is not a failure but a natural way for the brain to stay organized.

- | | | | | |
|-----|---|-------------|---|-------------|
| (1) | 1 | By contrast | 2 | Until then |
| | 3 | In fact | 4 | For example |

- | | | |
|-----|---|-------------------------------|
| (2) | 1 | it becomes perfectly clear |
| | 2 | it becomes harder to remember |
| | 3 | it turns into a new memory |
| | 4 | it is saved forever |

- | | | |
|-----|---|-----------------------------|
| (3) | 1 | tend to forget more quickly |
| | 2 | remember everything easily |
| | 3 | never need to study |
| | 4 | become much smarter |

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No.16 / ラクダの砂漠適応

How Camels Survive in the Desert

Camels are famous for living in hot, dry deserts where there is little water and food. They can travel for many days without drinking, which seems impossible for such a large animal. Many people believe that camels store water in their humps, but (1). In fact, the hump is full of fat, not water.

The fat in the hump is an important energy store. When food is hard to find, the camel's body can break down this fat for energy. The camel also saves water in clever ways. For example, (2). Its body produces very little sweat and only loses a small amount of water when it breathes.

Camels have other useful features too. Their long eyelashes keep sand out of their eyes, and their wide feet stop them from sinking into the soft sand. They can also close their nostrils to keep out blowing sand during desert storms. (3), the camel's body is perfectly suited to desert life. These adaptations have made it one of the most important animals for people living in dry regions.

- | | | | | |
|-----|---|--------------------|---|-------------------|
| (1) | 1 | this is well known | 2 | the hump is empty |
| | 3 | this is not true | 4 | they drink often |

- | | | |
|-----|---|---|
| (2) | 1 | it drinks all day long |
| | 2 | it sweats a great deal |
| | 3 | it keeps as much water inside as possible |
| | 4 | it eats only meat |

- | | | | | |
|-----|---|-------------|---|-------------|
| (3) | 1 | By contrast | 2 | In this way |
| | 3 | Until then | 4 | However |

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B

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No.17 / 鳥肌が立つ理由

Why We Get Goosebumps

When we feel cold or suddenly afraid, tiny bumps often appear on our skin. These are called goosebumps. They form when small muscles at the base of our body hairs tighten and pull the hairs upright. But why does (1)?

The answer goes back to our distant ancestors, who had much more body hair than we do. When an animal with thick fur is cold, raising its hair traps a layer of warm air close to the skin. When it is frightened, (2). Standing hair makes the animal look bigger and more dangerous to its enemies.

Humans no longer have thick fur, so goosebumps do not really keep us warm or protect us. (3), the reaction still happens because it is built into our bodies from long ago. Goosebumps are a small reminder of how closely we are connected to other animals.

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|-----|---|----------------------|---|-------------------|
| (1) | 1 | the skin turn blue | 2 | the hair fall out |
| | 3 | the body grow taller | 4 | our body do this |

- | | | |
|-----|---|---------------------------------------|
| (2) | 1 | the animal loses its fur |
| | 2 | the raised fur serves another purpose |
| | 3 | the animal feels nothing |
| | 4 | the fur turns white |

- | | | | | |
|-----|---|-------------|---|-------------|
| (3) | 1 | Even so | 2 | For example |
| | 3 | By contrast | 4 | Until then |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.18 / カメレオンの変色

How Chameleons Change Color

Many people think that chameleons change color mainly to hide from enemies by matching their background. This is partly true, but (1). Chameleons actually change color for several reasons, including to communicate with other chameleons and to control their body temperature.

A chameleon's skin contains special cells with tiny crystals inside. By changing the space between these crystals, (2). When the crystals are close together, the skin reflects one color; when they move apart, it reflects another. This is how a chameleon can turn from green to yellow or even bright red.

Color also sends messages. A male chameleon may turn bright colors to warn other males away or to attract a mate. A darker color can help it absorb more heat from the sun on a cold morning. (3), a chameleon's color is a kind of language as well as a way to stay comfortable.

- (1)
- 1 it explains everything
 - 2 it is not the whole story
 - 3 they never hide
 - 4 color does not change

- (2)
- 1 the skin becomes harder
 - 2 the chameleon goes blind
 - 3 the crystals disappear
 - 4 the animal changes the color it reflects

- (3)
- | | |
|---------------|---------------|
| 1 However | 2 By contrast |
| 3 In this way | 4 Until then |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.19 / 犬の優れた嗅覚

How Dogs Smell So Well

A dog's sense of smell is far more powerful than a human's. While people have about five million smell receptors in their noses, dogs have around three hundred million. (1), dogs can detect smells that are far too weak for humans to notice. This amazing ability shapes how dogs experience the world.

Dogs also smell in a special way. Unlike humans, a dog can breathe in and smell at the same time, without one action getting in the way of the other. In addition, each side of a dog's nose (2). This helps the dog tell which direction a smell is coming from.

Because of these abilities, people train dogs to do important jobs that depend on smell. Some dogs help find people who are lost in the mountains, while others (3). A few specially trained dogs can even smell certain illnesses in the human body, helping doctors find problems early.

(1)

1 By contrast

2 However

3 As a result

4 Until then

(2)

1 works on its own

2 stops breathing

3 smells nothing

4 closes completely

(3)

1 cannot be trained

2 dislike working

3 lose their sense of smell

4 search for dangerous items

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.20 / アリの採餌

How Ants Find Food

A single ant is very small and not very clever, but a colony of ants can find food and bring it home with surprising skill. The secret is teamwork and a clever use of smell. When an ant finds food, it does not keep the discovery to itself. Instead, (1).

As the ant walks back to the nest, it drops a special chemical on the ground. Other ants can smell this chemical and follow the trail straight to the food. (2), more and more ants use the same path, and each one adds more chemical as it passes. Soon, a strong, clear trail leads the whole colony to the food, and the ants carry it back piece by piece.

When the food is gone, the ants stop adding the chemical, and the smell slowly fades until the trail (3). After that, the ants no longer follow the old path and are free to search for food somewhere else. This simple system, with no leader giving orders, allows thousands of ants to work together and solve difficult problems as a single team.

- (1)
- | | |
|---|--|
| 1 | it eats everything quickly |
| 2 | it hides the food |
| 3 | it forgets the way home |
| 4 | it leaves a trail for others to follow |

- (2)
- | | | | |
|---|-------------|---|-------------|
| 1 | By contrast | 2 | However |
| 3 | Until then | 4 | As a result |

- (3)
- | | | | |
|---|-----------------|---|-----------------------|
| 1 | grows stronger | 2 | disappears completely |
| 3 | turns into food | 4 | leads them home |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.21 / フラミンゴが桃色の理由

Why Flamingos Are Pink

Flamingos are famous for their beautiful pink feathers, but baby flamingos are not pink at all. When they hatch, (1). Their feathers are gray or white, and they slowly turn pink only as the birds grow older. The reason for this color change is surprising: it comes from their food.

Flamingos eat small shrimp and tiny water plants called algae. These foods contain natural color substances that are red and orange. As a flamingo eats more and more of them over time, (2). The color substances build up in the bird's body and slowly turn its feathers pink.

This means that a flamingo's color depends on its diet. Flamingos that eat a lot of these foods become a deep pink, while those that eat less stay paler. (3), you can often tell how a flamingo has been eating just by looking at the brightness of its feathers.

- | | | | | |
|-----|---|---------------------|---|-----------------------|
| (1) | 1 | they are bright red | 2 | they are a dull color |
| | 3 | they cannot eat | 4 | they are fully grown |

- | | | |
|-----|---|------------------------------|
| (2) | 1 | it stops eating |
| | 2 | it becomes smaller |
| | 3 | its feathers begin to change |
| | 4 | it loses its color |

- | | | | | |
|-----|---|-------------|---|------------|
| (3) | 1 | In fact | 2 | However |
| | 3 | By contrast | 4 | Until then |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.22 / 虹が見える理由

Why We See Rainbows

A rainbow is one of the most beautiful sights in nature, appearing as a great arc of colors in the sky. Rainbows usually appear after rain, when the sun comes out again. (1), both sunlight and water drops are needed to create one. Without both together, a rainbow cannot form.

Sunlight may look white, but it is really a mix of many colors. When sunlight passes into a raindrop, the light bends, and (2). Each color bends by a slightly different amount, so the colors spread out and become separate. The light then reflects inside the drop and comes back out toward our eyes.

Because the colors leave the raindrops at different angles, we see them as separate bands stretched across the sky. This is why a rainbow always (3). The colors appear in the same order every time, from red on the outside to violet on the inside, no matter where in the world you see it.

- | | | | | |
|-----|---|-------------|---|----------------|
| (1) | 1 | By contrast | 2 | For example |
| | 3 | Until then | 4 | In other words |

- | | | |
|-----|---|---------------------------------|
| (2) | 1 | it splits into different colors |
| | 2 | it disappears at once |
| | 3 | it turns into rain |
| | 4 | it becomes hotter |

- | | | |
|-----|---|------------------------------------|
| (3) | 1 | looks completely different |
| | 2 | has only one color |
| | 3 | shows its colors in the same order |
| | 4 | appears at night |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.23 / 海が塩辛い理由

Why the Ocean Is Salty

Anyone who has swum in the sea knows that ocean water is salty, while the water in most rivers and lakes is not. Have you ever wondered where all this salt comes from? The answer is that (1). Most of the salt in the ocean has been carried there, little by little, by rivers.

As rain falls on rocks and soil, it slowly breaks them down and picks up tiny amounts of minerals, including salt. Rivers then carry these minerals down to the sea. The water itself does not stay in the ocean forever; (2). It evaporates into the air to form clouds, but the salt is left behind in the sea.

This process has continued for millions of years. Each year, rivers add a little more salt, and each year, water evaporates and leaves the salt behind. (3), the amount of salt in the ocean has slowly built up over an enormous length of time. The sea is salty because of a process that never stops.

- (1)
- 1 the salt is made by fish
 - 2 the ocean has always been empty
 - 3 the salt comes from the land
 - 4 rivers are very salty

- (2)
- 1 the salt also disappears
 - 2 only the water leaves
 - 3 nothing ever changes
 - 4 the rivers run dry

- (3)
- | | |
|--------------|---------------|
| 1 However | 2 By contrast |
| 3 Until then | 4 In this way |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.24 / 熱気球が浮く仕組み

How Hot Air Balloons Rise

A hot air balloon can carry people high into the sky using a very simple idea about air. The key is that hot air is lighter than cool air. When the air inside the balloon is heated, (1). Because it is lighter than the cooler air outside, the whole balloon begins to float upward.

To make the balloon rise, the pilot uses a strong flame to heat the air inside the large cloth bag. The more the air is heated, the lighter it becomes, so the balloon climbs higher. (2), to come back down, the pilot simply stops heating the air. As the air slowly cools, it grows heavier, and the balloon gently sinks.

Although the science is simple, flying a balloon takes skill. The pilot cannot steer the balloon left or right directly. Instead, (3). By moving the balloon up or down, the pilot finds winds blowing in different directions and uses them to travel the desired way.

- (1)
- 1 it becomes lighter
 - 2 it becomes much heavier
 - 3 it turns to water
 - 4 it disappears

- (2)
- | | |
|---------------------|--------------|
| 1 For example | 2 Until then |
| 3 On the other hand | 4 In fact |

- (3)
- 1 the balloon flies by itself
 - 2 the pilot uses a steering wheel
 - 3 the balloon never moves
 - 4 the pilot changes height to catch the wind

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.25 / 紅葉の理由

Why Leaves Change Color in Autumn

In autumn, the leaves of many trees change from green to bright shades of yellow, orange, and red. This happens because of changes in the weather and the amount of sunlight. During spring and summer, leaves are green because (1). A substance called chlorophyll, which the tree uses to make food from sunlight, is green and fills the leaves.

As autumn comes, the days grow shorter and cooler. The tree prepares to rest for the winter and stops making chlorophyll. As the green chlorophyll breaks down and fades, (2). These yellow and orange colors were in the leaf all along, but the strong green had hidden them.

Some trees also make a new red substance in their leaves in autumn, which adds bright red to the display. Soon after, the leaves fall to the ground, and the tree rests until spring. (3), the beautiful colors of autumn are really a sign that the trees are getting ready for winter.

- (1)
- 1 they have no color
 - 2 they are full of a green substance
 - 3 they are very cold
 - 4 they receive no light

- (2)
- 1 the leaf turns greener
 - 2 the leaf grows larger
 - 3 the tree dies at once
 - 4 other colors become visible

- (3)
- | | |
|---------------|--------------|
| 1 In this way | 2 However |
| 3 By contrast | 4 Until then |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.26 / 傍観者効果

The Bystander Effect

Imagine someone falls down on a busy street full of people. You might think that the more people there are, the more likely someone will help. (1), the opposite is often true. When many people are present, each person is actually less likely to help. Psychologists call this surprising fact the "bystander effect."

There are a few reasons for this. In a large group, each person assumes that (2). People also look around to see how others are reacting, and if no one else seems worried, they decide that the situation must not be serious. As a result, everyone waits, and no one acts.

Knowing about the bystander effect can help us act more wisely. If you ever need help in a crowd, experts say you should point to one person and ask them directly. When you do this, that person (3). Feeling personally responsible, they are much more likely to step forward and help.

- | | | |
|-----|--|-----------------------|
| (1) | 1 For example | 2 As a result |
| | 3 However | 4 In the same way |
| (2) | 1 someone else will help | 2 they alone can help |
| | 3 the person is fine | 4 help is not needed |
| (3) | 1 quickly walks away | |
| | 2 can no longer assume others will act | |
| | 3 becomes very angry | |
| | 4 ignores the request | |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.27 / 夢を見る理由

Why We Dream

Everyone dreams, even people who do not remember their dreams in the morning. Dreams happen mostly during a stage of sleep when the brain is very active. For a long time, scientists have asked a difficult question: (1). Although there is still no single answer, several interesting ideas have been suggested.

One popular idea is that dreaming helps the brain deal with memories. While we sleep, the brain sorts through what happened during the day, keeping important information and removing what is not needed. According to this idea, (2). The strange stories in our dreams may simply be a side effect of the brain organizing itself.

Another idea is that dreams allow us to practice dealing with difficult or frightening situations in a safe way. By facing fears in a dream, where the situation cannot cause any real harm, a person may feel better prepared in real life. (3), dreaming may be far more useful than it seems. Scientists continue to study sleep to understand what dreams really do.

- | | | | | |
|-----|---|-----------------------|---|---------------------|
| (1) | 1 | how we fall asleep | 2 | when night begins |
| | 3 | where dreams are sold | 4 | why we dream at all |

- | | | |
|-----|---|---------------------------------|
| (2) | 1 | dreams have no cause |
| | 2 | dreams are part of this process |
| | 3 | the brain stops working |
| | 4 | memories are lost forever |

- | | | | | |
|-----|---|-------------|---|-------------|
| (3) | 1 | However | 2 | By contrast |
| | 3 | In any case | 4 | Until then |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.28 / 第一印象の力

The Power of First Impressions

When we meet someone for the first time, we form an opinion of them very quickly—often within just a few seconds. Based on a person's face, voice, and clothes, (1). We decide whether we think they are friendly, honest, or smart, long before we really know them. This is the power of first impressions.

First impressions are powerful because they are hard to change. Once we believe something about a person, (2). If we have decided that someone is kind, we tend to notice their kind actions and explain away anything that does not fit. Even when later evidence suggests we were wrong, we often hold on to our first opinion.

First impressions are so strong and so lasting that they can shape important moments in life, such as job interviews or first dates. (3), it is worth remembering that a quick judgment is not always a fair one. Taking time to know someone can help us see who they really are.

- (1)
- 1 our brain makes a fast judgment
 - 2 we forget them at once
 - 3 we say nothing at all
 - 4 time stops completely

- (2)
- 1 we quickly forget it
 - 2 we change our minds easily
 - 3 we ask many questions
 - 4 we look for proof that we are right

- (3)
- | | |
|-------------------|---------------------|
| 1 By contrast | 2 On the other hand |
| 3 For this reason | 4 Until then |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.29 / 単純接触効果

Why We Like Familiar Things

Have you noticed that a song you disliked at first can become a favorite after you hear it several times? Psychologists have found that, in general, (1). The more often we see or hear something, the more we tend to like it, and this pattern is known as the "mere-exposure effect."

Why does this happen? One idea is about safety. Long ago, something unfamiliar might have been dangerous, so being careful around new things helped people survive. When we meet something many times without being harmed, (2). Our brain treats the familiar thing as safe, and this feeling of safety becomes a kind of liking.

The mere-exposure effect is used widely in the real world. Companies show their advertisements again and again so that people will come to feel familiar with their products and like them. (3), simply seeing a brand many times can make us more likely to choose it, even if we never think about the ads at all.

- (1)
- 1 people prefer new things
 - 2 familiarity makes us like things more
 - 3 music is always annoying
 - 4 we dislike repeated sounds

- (2)
- 1 we begin to feel comfortable with it
 - 2 we become more afraid
 - 3 we forget it completely
 - 4 we feel nothing at all

- (3)
- | | |
|--------------|---------------|
| 1 However | 2 By contrast |
| 3 Until then | 4 In this way |

2

B

次の英文を読み、その文意にそって(1)から(3)までの()に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

No.30 / 習慣ができる仕組み

How Habits Form

Brushing your teeth, checking your phone, biting your nails—these are all habits, actions we do automatically without much thought. Habits are useful because they (1). When an action becomes a habit, the brain no longer has to think hard about it, which saves time and energy for more important decisions.

Scientists have found that most habits follow a simple pattern with three parts. First comes a "cue," something that triggers the behavior, such as feeling bored. Next comes the routine, the action itself. (2), there is a reward—a good feeling that makes the brain want to repeat the action. Over time, this loop becomes stronger and harder to break.

Understanding this pattern can help people change bad habits. Experts say that the best way is not to fight the habit directly but to (3). By keeping the same cue and reward while replacing the routine with a healthier action, a person can slowly build a better habit in place of the old one.

- (1)
- 1 make us very tired
 - 2 take a long time
 - 3 free the brain from extra effort
 - 4 are always harmful

- (2)
- | | |
|---------------|-----------|
| 1 By contrast | 2 Finally |
| 3 Until then | 4 However |

- (3)
- 1 change just one part of the loop
 - 2 ignore the problem completely
 - 3 give up all habits
 - 4 repeat the bad action

解答一覧 | Answer Key

No.	トピック	(1)	(2)	(3)
1	Why Yawning Spreads	1	3	2
2	How Bees Talk	4	2	1
3	How Bats See with Sound	2	4	3
4	How Geckos Climb Walls	3	1	4
5	How Trees Share Food	1	4	2
6	How Birds Find Their Way	2	3	1
7	How Penguins Stay Warm	4	1	3
8	How Spider Silk Is So Strong	3	2	4
9	Why the Sky Is Blue	1	3	4
10	Why Ice Floats	2	4	1
11	How the Body Stays at the Right Temperature	3	1	2
12	The Placebo Effect	4	2	3
13	The Spotlight Effect	1	4	3
14	Why Music Moves Us	2	1	4
15	Why We Forget	3	2	1
16	How Camels Survive in the Desert	3	3	2
17	Why We Get Goosebumps	4	2	1
18	How Chameleons Change Color	2	4	3
19	How Dogs Smell So Well	3	1	4
20	How Ants Find Food	4	4	2
21	Why Flamingos Are Pink	2	3	1
22	Why We See Rainbows	4	1	3
23	Why the Ocean Is Salty	3	2	4
24	How Hot Air Balloons Rise	1	3	4
25	Why Leaves Change Color in Autumn	2	4	1
26	The Bystander Effect	3	1	2
27	Why We Dream	4	2	3
28	The Power of First Impressions	1	4	3
29	Why We Like Familiar Things	2	1	4
30	How Habits Form	3	2	1